Sampling for Uniform Observability in a State-Dependent Riccati Estimator

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Abstract— **In this paper we consider the continuous-discrete formulation of the State Dependent Riccati equation-based Estimator (SDRE) for nonlinear systems. In an SDRE, the nonlinear system equation is parameterized by a state transition matrix and this transition matrix is then used in a Riccati equation to propagate the error covariance. In this paper, we show that if the state-dependent transition matrix and the output matrix form an observable pair, then under certain sampling conditions, the SDRE would be uniformly observable and thus, the estimator error covariance will be bounded. The implication is that for certain nonlinear systems, the convergence of the SDRE can be established in advance. This result is illustrated using a practical example.**

I. INTRODUCTION

Motivation. State estimation is vital in many applications [1], [2] because it is often infeasible to obtain full-state measurement of a system. However, if the system is observable, then it is possible to fully reconstruct or estimate the system state from limited measurements [3]. For a deterministic linear time-invariant (LTI) system, a Luenberger observer is generally used to determine the unmeasured states [4]. If some noise is present in the system or in the measurement sensors, the Kalman filter provides the optimal state estimates [5]. For nonlinear systems, on the other hand, estimation techniques are generally sub-optimal [6]. Moreover, there are no guarantees, in advance, that the error between the estimated states and the actual state converges in finite time to zero.

Gaps in Literature. The extended Kalman filter (EKF) is the most commonly used estimator for nonlinear systems. The EKF differs from the standard Kalman filter only in that the system is linearized using a first-order approximation at specified time intervals. However, the linearization step is computationally expensive to implement. Furthermore, the local linearization may yield a Jacobian that makes the system unobservable about the local point [7]. Hence, stability analysis of nonlinear filters typically assumes the linearization would always maintain observability and, consequently, the boundedness of the estimation error covariance [8]–[10]. Another widely used variant of the Kalman filter, the unscented Kalman filter (UKF) does not require the computation of the Jacobian in its implementation [11], [12]. But again, stability proofs for the filter require the Jacobian to satisfy conditions that are difficult, in practice, to establish [13]–[15]. Additionally, in the absence of noise, the UKF, as an observer, does not converge [16].

In [7], another technique for nonlinear state estimation, the State Dependent Riccati equation-based Estimator (SDRE), is proposed. The SDRE has the same architecture as the extended Kalman filter but instead of the first order linearization, the governing equation is parametrized by a state-dependent transition matrix. This transition matrix is then used in a Riccati equation to update the error covariance. The SDRE has been formulated both as a continuous and discrete-time estimator [17]–[20]. In continuous time, it has been shown that if the state dependent transition matrix and the output matrix form a detectable pair for all states, the resulting error covariance is bounded and the SDRE is an exponential observer, that is, the estimation error exponentially converges to zero [18]. This implies that the stability of the observer can be established in advance. Meanwhile, in [20] it was shown that for a purely discrete time system, if the error covariance is bounded, under certain design conditions, the SDRE estimate converges.

Contribution. In practice, the dynamics of most systems are continuous, whereas measurements may only be taken in discrete time steps. Hence, in this paper, we fill the gap in the stability analysis for a continuous-discrete SDRE formulation—that is, one in which the original governing equations are continuous, but the output equation is discrete. First, we distinguish between pointwise observability, where the output matrix and the state transition matrix form an observable pair, and uniform observability, where the output matrix and a series of state transition matrices form an observable set. Second, we demonstrate that the pointwise observability rank test, used in [21], [22], does not generally translate to uniformly observability, and hence, cannot be used to assess the continuous-discrete SDRE stability. Third, we give conditions on sampling under which uniformly observability, and thus the boundedness of the error covariance, can be established.

Outline. The paper is organized as follows. In Sec. II, we define basic concepts used in the rest of the paper. Sec. III includes a description of the need for uniform observability and highlights how it differs from local observability. Main results of the paper, on uniform observability and boundedness, are given in Sec. IV. In Sec. V, we provide an illustrative example to demonstrate SDRE estimation for a uniformly observable system. We conclude the paper in Sec. VI.

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II. Preliminaries

Consider the system

$$\dot{x} = f(x) \tag{1a}$$
$$y = Cx \tag{1b}$$

where $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$ and $C \in \mathbb{R}^{m\times n}$.

A. Linear System Observability

If $f(x)$ is LTI, that is, $f(x) = Ax$ for some constant $A \in \mathbb{R}^{n\times n}$, then the following notion of observability holds.

Definition 1 (LTI Observability, e.g. [23])**.** *For some constant $A \in \mathbb{R}^{n\times n}$ and $C \in \mathbb{R}^{m\times n}$, the pair (A, C) is observable if*

1) *the observability matrix*

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \tag{2}$$

has rank n;

2)
$$\mathcal{O}^T\mathcal{O} = \sum_{i=0}^{n-1} A^{i^T} C^T C A^i \succ 0. \tag{3}$$

Rather than having a continuous output, suppose we sample the system at every time t_i, $i = 1, 2, \ldots$, then we have

$$x_{i+1} = \Phi(t_{i+1}, t_i)\, x_i \tag{4a}$$
$$y_i = Cx_i \tag{4b}$$

where $x_i = x(t_i)$, $\Phi(t_{i+1}, t_i) = e^{A\Delta t_i}$, and $\Delta t_i = t_{i+1} - t_i$. If the continuous-time system is observable, a suitable Δt_i preserves observability for the discrete-time system.

Definition 2 (e.g. [24])**.** *The sampling period Δt_i is* non-pathological *if the observability of (A, C) implies that $(\Phi(t_{i+1}, t_i), C)$ is also observable.*

Remark 1. (i) For constant sampling, $\Delta t_i = \Delta_s$ is non-pathological provided A has no eigenvalue pair $\{\lambda, \lambda'\}$ such that [24]

$$(\lambda - \lambda')\Delta_s = 2\pi j z \text{ for } z = \pm 1, \pm 2, \ldots \tag{5}$$

(ii) When sampling is non-uniform, sampling schedules for Δt_i that preserve observability are given in [24]–[26].

B. Local Observability for Quasi-Linear-Parameter-Varying (Q-LPV) Representation of a Nonlinear System

For a nonlinear $f(x)$, we can parameterize (1a) as a quasi-linear-parameter-varying system (see [27], [28]), that is, $f(x) = A(x(t))\, x$ [1]. Assuming that over the interval τ, $A(x(t))$ is constant, (1a) can be written as:

$$\dot{x} = A_k x(t) \text{ for } t \in [k\tau \;\; (k+1)\tau) \tag{6}$$

where $A_k \in \mathbb{R}^{n\times n}$ is the value of $A(x(t))$ at time $k\tau$: $A_k = A(x(k\tau))$. For this Q-LPV system, the notion of observabililty at $k\tau$ follows from that of an LTI system.

Definition 3 (Local Observability)**.** *The pair (A_k, C) is locally observable at the k^{th} time step if*

$$\mathcal{O}_k = \begin{bmatrix} C \\ CA_k \\ \vdots \\ CA_k^{n-1} \end{bmatrix} \tag{7}$$

is full rank, and hence $\mathcal{O}_k^T\mathcal{O}_k = \sum_{l=0}^{n-1} A_k^{l^T} C^T C A_k^l \succ 0$. The system is continuously locally observable (CLO) *if, for any given time step size τ, $\mathcal{O}_k$ is full rank $\forall$ k.*

Suppose the nonlinear system is sampled N times every interval τ. Then (4) holds with $\Phi(t_{i+1}, t_i) = e^{A_k\Delta t_i}$ and $k\tau = \sum_{i=1}^{kN} \Delta t_i$, $k = 1, 2, \ldots$. Since the parameterization of the system varies with time, the preservation of observability, in general, only applies locally. In other words, for non-pathological Δt_i the observability of (A_k, C) only necessarily implies that $(\Phi(t_{i+1}, t_i), C), kN \le i \le (k+1)N$ is observable:

$$\begin{aligned}\mathcal{O}_k^T\mathcal{O}_k &= \sum_{l=0}^{n-1} A_k^{l^T} C^T C A_k^l \succ 0 \\ \implies &\sum_{i=kN}^{kN+n-1} \Phi^T(t_i, t_{kN}) C^T C \Phi(t_i, t_{kN}) \succ 0.\end{aligned} \tag{8}$$

We term a sampling routine where $N = 1$, that is, the system is sampled only at times $k\tau$, $k = 1, 2, \ldots$, as *frozen-in-time* sampling.

C. Uniform Observability for Q-LPV System

Generally, the notion of observability for the Q-LPV system applies to any arbitrary sampling instances, t_i, $i = 1, 2, \ldots$, and not necessarily within the interval $[k\tau \;\; (k+1)\tau)$ where the parameterization of $A(x(t))$ is frozen-in-time. Let $\mathcal{S} = \{i : i \in \mathbb{Z}_+\}$. We define the state transition matrix for *any* successive integer pair $a, b \in \mathcal{S}$, $b > a$, as

$$\Phi_{b|a} = \Phi_{b-1} \cdots \Phi_a \tag{9}$$

where $\Phi_i = \Phi(t_{i+1}, t_i)$ $\forall i$. Note that for $b = a$, $\Phi_{b|a} = I_n$ and for $b < a$, $\Phi_{b|a}$ is not defined. The transition matrix Φ_i can be regarded as the discrete-time state evolution matrix. Thus, for the pair (Φ_i, C), we have a discrete-time observability Gramian defined as:

$$\mathcal{W}_{b,a} \triangleq \sum_{i=a}^{b} \Phi_{i|a}^T C^T C \Phi_{i|a}. \tag{10}$$

Definition 4 (Uniform Complete Observability (UCO) [29])**.** *The pair (Φ_i, C) is uniformly completely observable if $\forall i >$*

[1] The parameterization can be properly expressed as

$$\dot{x} = A(\theta)x(t)$$
$$\theta = \eta(x(t)).$$

Since the parameter θ is a function of $x(t)$, we may also write $A(\theta)$ as $A(x(t))$. Note that the parametrization $A(\theta)$ is not unique (see [20]).

0, *there exist positive constants* $\alpha_2 > \alpha_1 > 0$ *and an integer* $h > 0$ *such that the observability Gramian* $W_{i+h-1,i}$ *satisfies*

$$\alpha_1 I_n \preceq \mathcal{W}_{i+h-1,i} \preceq \alpha_2 I_n. \tag{11}$$

Remark 2. (i) Note that if the discrete time-varying observability matrix

$$\mathcal{O}_{n,i}^d = \begin{bmatrix} C \\ C\Phi_i \\ \vdots \\ C\Phi_{i+n|i} \end{bmatrix} \tag{12}$$

has rank n, the lower bound in (11) is satisfied.

(ii) The lower bound on the observability Gramian implies its invertibility; hence, the system's state at any time t_i can be reconstructed by a least squares approach (see [30]).

III. Problem Description

Suppose we have the system

$$\dot{x} = f(x) + w \qquad w(t) \sim \mathcal{N}(0, W(t)) \tag{13a}$$
$$y_i = Cx_i + v_i \qquad v_i \sim \mathcal{N}(0, V_i) \tag{13b}$$

with an estimator defined as follows:

$$\dot{\hat{x}} = f(\hat{x}) + w \tag{14a}$$
$$\hat{y}_i = C\hat{x}_{i|i} + v_i \tag{14b}$$

where

$$\begin{aligned} &\hat{x}_{i|i} = \hat{x}(t_i),\ \hat{x}_{i+1|i} = \hat{x}(t_{i+1}), \\ &\hat{x}_{i+1|i+1} = \hat{x}_{i+1|i} + L_i(y_i - \hat{y}_i). \end{aligned} \tag{15}$$

In (15), L_i is the Kalman filter gain given by

$$L_i = P_{i+1|i}C^T\left(CP_{i+1|i}C^T + V_i\right)^{-1}, \tag{16}$$

where $P_{i+1|i}$ is obtained using the state-dependent Riccati equation (SDRE):

$$\dot{P}(t) = A_iP(t) + P(t)A_i^T + W(t) \text{ for } t \in [t_i\ t_{i+1}), \tag{17a}$$
$$P_{i|i} = P(t_i), \quad P_{i+1|i} = P(t_{i+1}), \tag{17b}$$
$$P_{i+1|i+1} = (I - L_iC)\,P_{i+1|i}, \tag{17c}$$

and $A_i = A(\hat{x}_{i|i})$.

In classical results, the existence of a bounded solution for the error covariance $P_{i+1|i}$ relies on UCO of the system [30], [31]. For an LTI system, UCO amounts only to satisfying the observability rank test in Definition 1. In [21], [22], it is suggested (without proof) that for the continuous-discrete SDRE, CLO at the frozen-in-time samples would ensure a bounded $P_{i+1|i}$. However, as we show, in the following counter-example, CLO does not necessarily translate to UCO, and hence, is not generally sufficient or necessary for assessing stability of the SDRE.

Counterexample. Consider the system with

$$A(x(t)) = \begin{bmatrix} 0 & f_2(x(t)) & f_3(x(t)) \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \tag{18}$$

where f_2 and f_3 are functions of $x(t)$ and $C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$. Assume we have a frozen-in-time sampling routine at $t = k$, for $k = 1, 2, \ldots$, with a unit time interval. Then,

$$A_k = \begin{bmatrix} 0 & f_{2,k} & f_{3,k} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \Phi_k = \begin{bmatrix} 1 & f_{2,k} & f_{3,k} + \frac{f_{2,k}}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}. \tag{19}$$

where $f_{2,k}$ and $f_{3,k}$ are the values of their respective functions at $t = k$. If $f_2(x(t)) > 0\ \forall t$, it can be verified that $\mathcal{O}_k$ (in (7)) and $\mathcal{O}_k^d$ ($\mathcal{O}_k$ where A_k is replaced with Φ_k) are always full rank $\forall k$. However $\mathcal{O}_{n,k}^d$ (in (12)) is not full rank if $f_{3,k} = -\frac{f_{2,k}}{2}$ and $f_{3,k+1} = -\frac{3f_{2,k+1}}{2}$.

In view of the above clarification, our aim, in the following section, is to specify a sufficient sampling condition that guarantees that CLO results in UCO.

IV. Results

In this section, we present results as to what sampling condition enables a continuously locally observable (CLO) system to be also uniformly observable. We also show that the covariance matrix of the continuous discrete SDRE is bounded.

Assumption 1 (System Boundedness): There exist positive constants $\bar{a}, \bar{c}$ such that $\forall t$

$$\|A(x(t))\| \leq \bar{a} \tag{20}$$
$$\|C\| \leq \bar{c}. \tag{21}$$

Assumption 2 (Filter Boundedness): There exists positive constants $\bar{w}, \underline{v}, \bar{v}$, such that,

$$W(t) \leq \bar{w}I_n\ \forall t \tag{22a}$$
$$0 < \underline{v}I_m \leq V_i \leq \bar{v}I_m, \text{ for } i = 1, 2, \ldots. \tag{22b}$$

A. Sampling for UCO in Continuous-Discrete SDRE

Theorem 1. *Let the system* $(A(x(t)), C)$ *be continuously locally observable* $\forall t$. *Also, assume* $A(x(t)) \in \mathbb{R}^{n\times n}$ *is constant over every time interval* τ. *If the system is sampled at times* t_i, $i = 1, 2, \ldots$, *and there are* N *samples within each interval* τ, *then the pair* $(\Phi(t_{i+1}, t_i), C)$, *where* $\Phi(t_{i+1}, t_i) = e^{A(x(t_i))\Delta t_i}$ *is uniformly completely observable* $\forall i$ *provided* $N > n - 1$ *and* $\Delta t_i = t_{i+1} - t_i$ *is non-pathological.*

Proof. Let $a \in \mathcal{S}$. Define

$$N_a \triangleq \begin{cases} a + N & \text{if } (a \bmod N) = 0, \\ a + 2N - (a \bmod N) & \text{else.} \end{cases} \tag{23}$$

Let $b \in \mathcal{S}$ and $b \geq N_a$; the observability gramian $W_{b,a}$ is equal to

$$\mathcal{W}_{b,a} = \sum_{i=a}^{b} \Phi_{i|a}^T C^T C \Phi_{i|a} \tag{24}$$

$$= \sum_{i=a}^{N_a-N-1} \Phi_{i|a}^T C^T C \Phi_{i|a} + \sum_{i=N_a-N}^{N_a-1} \Phi_{i|a}^T C^T C \Phi_{i|a} + \sum_{i=N_a}^{b} \Phi_{i|a}^T C^T C \Phi_{i|a}$$

$$\succeq \sum_{i=N_a-N}^{N_a-1} \Phi_{i|a}^T C^T C \Phi_{i|a} \tag{25}$$

$$\succeq \sum_{i=N_a-N}^{N_a-N+n-1} \Phi_{i|a}^T C^T C \Phi_{i|a} \tag{26}$$

$$= \Phi_a^T \cdots \Phi_{N_a-N}^T \Big(C^T C \tag{27}$$

$$+ \Phi_{N_a-N+1}^T C^T C \Phi_{N_a-N+1} + \cdots + \Phi_{N_a-N+1}^T \cdots \Phi_{N_a-N+n-1}^T C^T C \Phi_{N_a-N+n-1} \cdots \Phi_{N_a-N+1} \Big) \Phi_{N_a-N} \cdots \Phi_a \tag{28}$$

$$= \Phi_{N_a-N|a}^T \times \tag{29}$$

$$\left(\sum_{i=N_a-N}^{N_a-N+n-1} \Phi_{i|N_a-N}^T C^T C \Phi_{i|N_a-N} \right) \Phi_{N_a-N|a}. \tag{30}$$

Equation (25) holds because $\Phi_{i|a}^T C^T C \Phi_{i|a} \succeq 0 \ \forall i \geq a$. Also, (26) applies since $N > n - 1$. Note that there is some $k \in \mathbb{Z}^*$ such that $kN = N_a - N$. From Definition 2, since Δt_i is non-pathological, the observability of $(A(x(t)), C)$ for $t \in [k\tau \ (k+1)\tau)$ implies that

$$\sum_{i=N_a-N}^{N_a-N+n-1} \Phi_{i|N_a-N}^T C^T C \Phi_{i|N_a-N} \succ 0. \tag{31}$$

Finally, because $\Phi_{N_a-N|a}$ is full rank, (30) is positive definite and we have

$$\mathcal{W}_{b,a} = \sum_{i=a}^{b} \Phi_{i|a}^T C^T C \Phi_{i|a} \succeq \alpha_1 I_n \succ 0 \tag{32}$$

where α_1 is some positive constant. Moreover, given the boundedness of $A(x(t))$ and C, there exists an upper bound constant α_2 such that

$$0 \preceq \alpha_1 I_n \preceq \mathcal{W}_{b,a} \preceq \alpha_2 I_n. \tag{33}$$

□

B. Boundedness of the SDRE Covariance Matrix

Theorem 2. *Under Assumptions 1 and 2, the state prediction error covariance matrix $P_{i+1|i}$ is upper bounded by $\bar{P}_{i+1|i} = \Phi_i^T P_{i|i} \Phi_i + \bar{W}_i$ where $\bar{W}_i = \frac{\bar{w}}{2\bar{a}}(e^{2\bar{a}\Delta t_i} - 1) I_n$.*

Proof. The solution to the Lyapunov equation of (17b) is given by:

$$P_{i+1|i} = \Phi_i P_{i|i} \Phi_i^T + \int_{t_i}^{t_{i+1}} \Phi(t_{i+1}, t) W(t) \Phi^T(t_{i+1}, t)\, \mathrm{d}t \tag{34}$$

where $\Phi_i = \Phi(t_{i+1}, t_i)$. Hence we only need to show that $\int_{t_i}^{t_{i+1}} \Phi(t_{i+1}, t) W(t) \Phi^T(t_{i+1}, t)\, \mathrm{d}t$ has an upper bound for each i. From (22a), we have that

$$\int_{t_i}^{t_{i+1}} \Phi(t_{i+1}, t) W(t) \Phi^T(t_{i+1}, t)\, \mathrm{d}t \preceq \bar{w} \int_{t_i}^{t_{i+1}} \Phi(t_{i+1}, t) \Phi^T(t_{i+1}, t)\, \mathrm{d}t \tag{35}$$

Now $\Phi(t_{i+1}, t) = e^{A_i(t_{i+1}-t)}$ where $kN \leq i \leq (k+1)N$ for some integer k and $A_i = A(x(t))$for $t \in [k\tau \ (k+1)\tau)$. Since $A(x(t))$ is bounded, we have that

$$\int_{t_i}^{t_{i+1}} \Phi(t_{i+1}, t) W(t) \Phi^T(t_{i+1}, t)\, \mathrm{d}t \preceq \bar{w} \int_{t_i}^{t_{i+1}} e^{A_i(t_{i+1}-t)} e^{A_i^T(t_{i+1}-t)} \mathrm{d}t \preceq \bar{w} \int_{t_i}^{t_{i+1}} e^{2\bar{a}(t_{i+1}-t)} I_n \mathrm{d}t = \frac{\bar{w}}{2\bar{a}}(e^{2\bar{a}\Delta t_i} - 1) I_n. \tag{36}$$

.

□

Corollary. *Under Assumptions 1 and 2, if the pair (Φ_i, C) is UCO, $\bar{P}_{i+1|i}$, $P_{i|i}$ and the Kalman gain L_i are bounded.*

Proof. The proof is as given in [30].

V. Example

A. 1-Dimensional Heat Transfer

To demonstrate the SDRE filter, we consider heat transfer through a rod, shown in Fig. 1, in which the thermal conductivity and specific heat are temperature dependent. The rod is divided into four control volumes of equal mass, and conductive heat transfer between volumes is modeled using a thermal resistance network. A known disturbance heat load $\dot{Q}(t)$ is applied to the left end of the rod, and the right end is adiabatic (i.e. there is no heat transfer through the right end). A temperature sensor is placed at control volume 1 for output measurement, and it is assumed that the heat load is known to the SDRE filter.

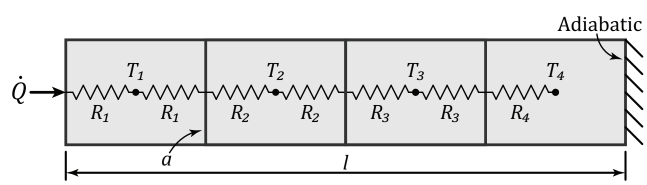

Fig. 1: 1-dimensional heat transfer through a rod of length l and cross-sectional area a modeled using four control volumes.

Equations (37a)-(37d) are the thermodynamic equations for the four control volumes. The thermal conductance between adjacent control volumes, $h_{j,j+1}$, is defined in (38). The thermal conductivity κ_j and the specific heat $c_{p,j}$ are functions of absolute temperature, defined in (39) and (40), respectively. These functions are arbitrary and do not represent the thermal properties of any particular material.

$$
\begin{aligned}
mc_{p,1}\dot{T}_1 &= h_{1,2}\left(T_2 - T_1\right) + \dot{Q} && \text{(37a)}\\
mc_{p,2}\dot{T}_2 &= h_{1,2}\left(T_1 - T_2\right) + h_{2,3}\left(T_3 - T_2\right) && \text{(37b)}\\
mc_{p,3}\dot{T}_3 &= h_{2,3}\left(T_2 - T_3\right) + h_{3,4}\left(T_4 - T_3\right) && \text{(37c)}\\
mc_{p,4}\dot{T}_4 &= h_{3,4}\left(T_3 - T_4\right) && \text{(37d)}
\end{aligned}
$$

$$h_{j,j+1} = \left(R_j + R_{j+1}\right)^{-1} = \left(\frac{l}{8a\kappa_j} + \frac{l}{8a\kappa_{j+1}}\right)^{-1} \tag{38}$$

$$\kappa_j(T_j) = \left(\frac{T_j}{270}\right)^3 \times 170 \left[\frac{\text{W}}{\text{m-K}}\right] \tag{39}$$

$$c_{p,j}(T_j) = \left(\frac{T_j}{270}\right)^{-3} \times 900 \left[\frac{\text{J}}{\text{kg-K}}\right] \tag{40}$$

Let

$$
\begin{aligned}
M(x(t)) &= m\begin{bmatrix} c_{p,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & c_{p,4}\end{bmatrix}\\
\Gamma(x(t)) &= \begin{bmatrix} -h_{1,2} & h_{1,2} & 0 & 0 \\ h_{1,2} & -h_{1,2}-h_{2,3} & h_{2,3} & 0 \\ 0 & h_{2,3} & -h_{2,3}-h_{3,4} & h_{3,4} \\ 0 & 0 & h_{3,4} & -h_{3,4}\end{bmatrix}.
\end{aligned} \tag{41}
$$

Then (37a)-(37c) may be re-written as a Q-LPV system given by

$$
\begin{aligned}
\dot{x}(t) &= A(x(t))x(t) + B(x(t))\dot{Q} + w(t) && \text{(42a)}\\
y_i &= Cx_i + v_i && \text{(42b)}
\end{aligned}
$$

where

$$
\begin{aligned}
x(t) &= \begin{bmatrix} T_1 & T_2 & T_3 & T_4 \end{bmatrix}^\top && \text{(43a)}\\
A(x(t)) &= M^{-1}(x(t))\Gamma(x(t)) && \text{(43b)}\\
B(x(t)) &= \begin{bmatrix} \frac{1}{mc_{p,1}} & 0 & 0 & 0 \end{bmatrix}^\top && \text{(43c)}\\
C &= \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}. && \text{(43d)}
\end{aligned}
$$

The signal $w(t)$ is Gaussian process noise with covariance $W = 0.5I_4$ applied to the system (simulation) to account for unknown disturbances and v_i is measurement noise v_i with variance $V = 1$.

The SDRE filter is formulated using the method described in Section III with $N = 4$ samples taken at a constant sample rate within each interval of $\tau = 1$ s. Samples are taken from a simulation of the 4-state system; the simulated temperatures are denoted as T_j, while the state estimates are denoted as $\hat{x}_j$. Various parameters required to recreate the simulation are included in Table I.

TABLE I: Model and SDRE filter parameters

Parameter	Value
l	0.15 m
a	1 cm^2
m	10.2 g
τ	1 s
Δt_i	0.25 s
N	4
$T(0)$	0°C
$\hat{x}(0)$	-20°C
$P(0)$	$300I_3$

B. Proof of Observability

It may be verified that for any time step k, the local observability matrix $\mathcal{O}_k$ of (7) is lower triangular and hence full rank. Moreover, $A_k = M_k^{-1}\Gamma_k$ is similar [32] to a symmetric matrix since Γ_k is symmetric and M_k^{-1} is a diagonal positive definite matrix. Therefore, the eigenvalues of A_k $\forall k$ are real and, as such, any constant sampling Δt_i is non-pathological. Consequently, by Theorem 1, pair $\left(\Phi\left(t_{i+1}, t_i\right), C\right)$ is uniformly completely observable $\forall i$ for $N > 3$.

C. Case Study

In the following case study, a heat load of 25 W is applied to the left end of the rod for 80 seconds. Then the heat load is removed, and the rod returns to a steady-state equilibrium. The initial state estimate, $\hat{x}_{1,2,3,4}(0) = -20$°C, is deliberately chosen to be far from the initial temperature of the simulation, $T_{1,2,3,4}(0) = 0$°C, so that convergence can be clearly demonstrated. Simulation results are compared to the corresponding state estimates in Fig. 2. With a measurement of T_1, the temperature in control volume 1, the estimate

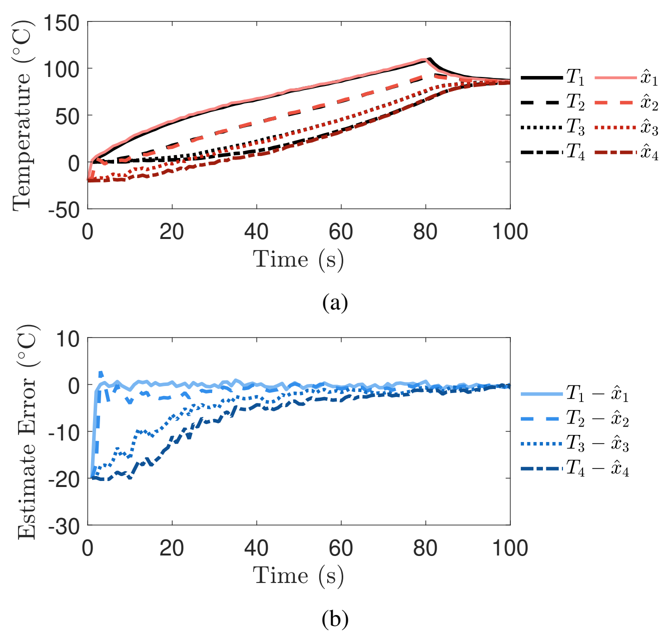

Fig. 2: a) Simulated (black) and estimated (red) temperatures of control volumes 1, 2, 3, and 4; b) Estimate errors for the four control volumes.

for this control volume $\hat{x}_1$ converges very quickly, and the estimates for the other two control volumes converge to within 2°C after 75 seconds despite the nonlinear dynamics of the rod. After the state estimates converge, the absolute values of the estimate errors remain less than 2°C, which can be explained by the measurement noise.

VI. Conclusion and Future Work

This paper has shown that for a discretely sampled system, local observability does not translate to uniform observability which is required, in classical results, for the boundedness of a SDRE estimator. We have thus proposed a minimum sampling rate under which local observability does implies uniform observability. An SDRE estimation scheme with such sampling routine was then applied to a heat conduction example. Future work would explore if there are non-trivial classes of systems for which local observability implies uniform observability without the sampling rate requirement.

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